



Aryl Hydrocarbon Receptor Signaling in Aging: A Key Modulator of Brain Homeostasis

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Abstract

Aging results from the gradual accumulation of molecular and cellular damage, and it is a risk factor for neurodegenerative disease. The aryl hydrocarbon receptor (AhR) is increasingly recognized as a key in the regulatory molecular mechanisms that drive aging. AhR is a cytoplasmic receptor activated by diverse endogenous and exogenous ligands. It modulates cell proliferation and differentiation in development and the adult brain. Besides, AhR participates in multiple physiological mechanisms that become dysregulated with age, including neuroinflammation, gut dysbiosis, stem cell maintenance, genomic stability, lipid metabolism, and mitochondrial function, contributing to the onset of neurodegeneration. These interconnected physiological processes lead to cumulative damage that ultimately impairs physical and cognitive function in older individuals. Activation of AhR by selective ligands has been shown to dampen neuroinflammatory responses and support cognitive integrity, underscoring its regulatory role in central nervous system (CNS) homeostasis. This review plumbs current evidence on how AhR influences the molecular mechanisms of aging and explores its potential as a therapeutic target to preserve brain health and mitigate age-related neurological decline.

Keywords AhR · Neuroinflammation · Gut microbiota · Dysbiosis and oxidative stress

Introduction

Aging results from the accumulation of molecular and cellular damage, leading to the deterioration of physical and mental capacities and an increased risk of illnesses [1]. Aging is the primary risk factor for many neurodegenerative diseases [2]. Furthermore, it is now recognized that peripheral organ dysfunction increases the incidence of age-related brain diseases. For example, diabetes and metabolic

syndrome increase the incidence of stroke and dementia, and chronic obstructive pulmonary disease and renal dysfunction are strongly associated with cognitive dysfunction and dementia [3].

In recent years, twelve hallmarks of aging have been identified, categorized as primary processes, compensatory mechanisms, and accumulated damage. Primary processes include proteostasis loss, genomic instability, epigenetic alterations, telomere shortening, and

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macroautophagic dysfunction. Compensatory or antagonistic processes involve cellular senescence, mitochondrial dysfunction, and dysregulation of nutrient-sensing systems. Lastly, processes related to accumulated damage include dysbiosis, altered intercellular communication, stem cell depletion, and chronic inflammation [4]. Each of these processes at the CNS has been associated with the development of common neurological disorders associated with aging, such as stroke and neurodegenerative diseases, including Alzheimer's disease (AD), Parkinson's disease (PD), and amyotrophic lateral sclerosis (ALS), among others, making them potential therapeutic targets [2, 5, 6]. Some examples of this relationship include altered intercellular communication characterized by chronic, low-grade inflammation known as inflammaging, and disruption of blood–brain barrier (BBB) integrity, which occurs during aging even in the absence of pathologies

and results from primary and compensatory processes. These changes lead to loss of homeostasis due to structural alterations and ultimately contribute to age-related diseases [7]. Among the various cellular and molecular changes in aging, AhR involvement has been identified in most primary, compensatory, and damage-associated processes. These changes lead to loss of homeostasis due to structural alterations and ultimately contribute to age-related diseases.

The activation of AhR is associated with various cellular and molecular processes that characterize aging, contribute to or mitigate CNS deterioration, and increase the risk of age-related disorders. In this review, we show current evidence on how AhR influences each molecular hallmark of aging and discuss its potential as a therapeutic target for preserving brain health and preventing or mitigating age-related diseases (Fig. 1).

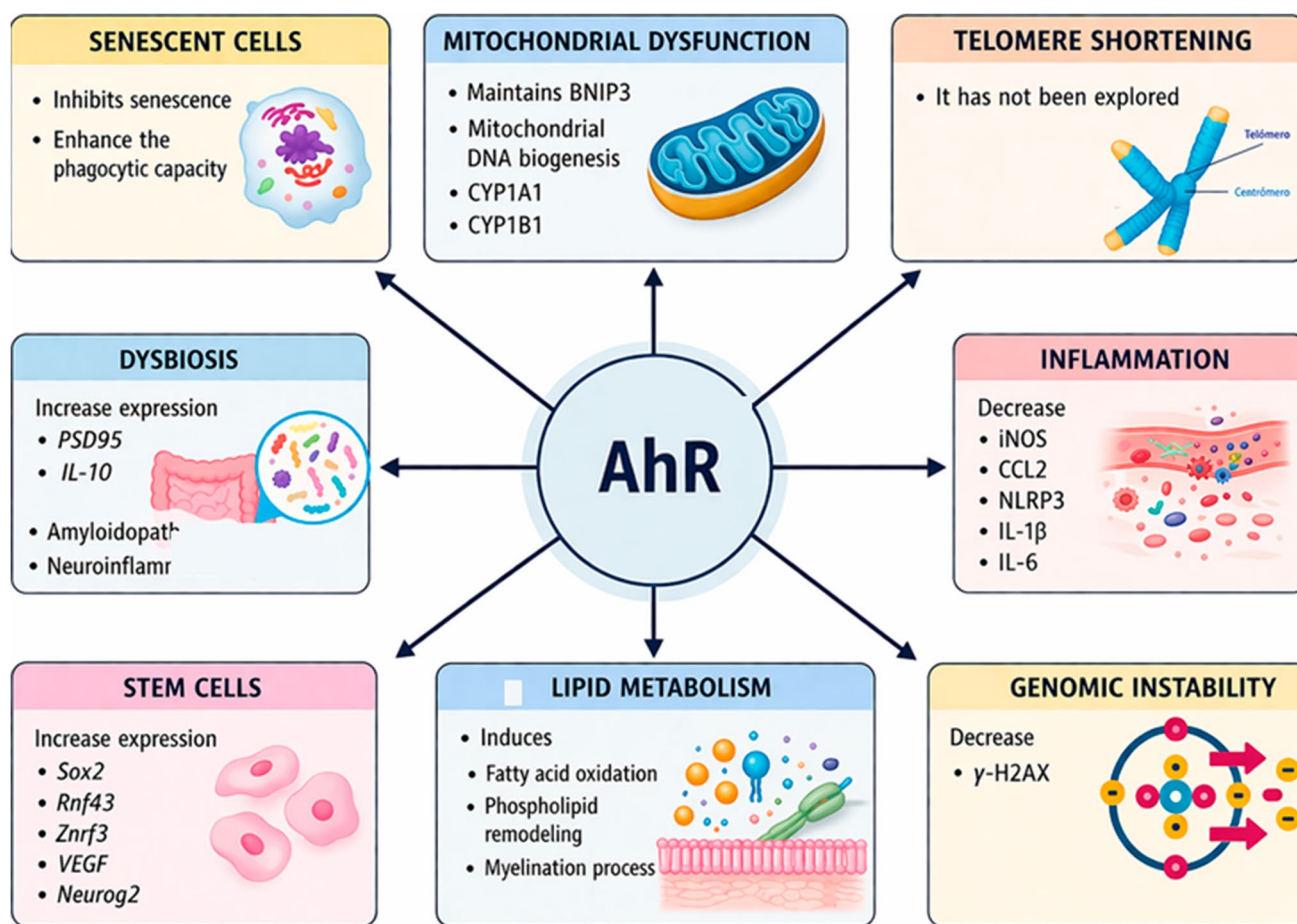


Fig. 1 Activation of the AhR and its involvement in aging hallmarks and neurodegenerative disease risk. The activation of AhR has been associated with various cellular and molecular processes that characterize aging. These include chronic inflammation, genomic instability, increased oxidative stress, alterations in lipid metabolism, mitochondrial dysfunction, cellular senescence, intestinal dysbiosis, and

impaired stem cell maintenance. Although the role of AhR in telomere shortening has not yet been thoroughly explored, its involvement in the other pathways suggests a systemic influence on the loss of homeostasis. These effects, mediated directly or indirectly by AhR activation, contribute to CNS deterioration and increase the risk of age-related diseases

The Aryl Hydrocarbon Receptor

The AhR is a member of the basic helix-loop-helix-ARNT-SIM (bHLH-PAS) transcription factor superfamily and is known for its activation by ligands [8]. It is recognized for its high affinity for 2,3,7,8-tetrachlorodibenzo-p-dioxin (TCDD) and other highly toxic and persistent halogenated organic pollutants, which initially led to its study in toxicology [9]. However, AhR also has several physiological functions, and its activation by diverse molecules has become an area of pharmacological interest. Among AhR agonists are dietary polyphenols, indoles, other tryptophan metabolites, commonly used drugs, and microbiota-derived molecules [10–17].

In the absence of a ligand, AhR remains inactive in the cytoplasm, forming a protein complex with Hsp90, ARA9, Src, and p23, which retains it in the cytoplasm and prevents its degradation [18, 19]. Upon ligand binding, a conformational change is induced, exposing the nuclear localization signal (NLS). This signal is recognized by importin- β , facilitating the translocation of the entire complex (AhR-ligand-chaperone proteins) into the nucleus. Within the nucleus, AhR dissociates from its chaperone proteins and dimerizes with the AhR nuclear translocator (ARNT), forming the AhR-ARNT heterodimer. This heterodimer then binds to AhR response elements (AHREs), regulating the expression of target genes [18, 20, 21] (Fig. 2).

AhR-induced protein expression plays multifaceted roles that extend beyond xenobiotic metabolism, regulating key processes such as cellular proliferation and differentiation, the maintenance of intestinal and BBB integrity, and the development of the vascular, immune, and CNS systems. Furthermore, AhR mediates intercellular communication via the release of tryptophan-derived metabolites [22–24].

AhR activity is linked to cellular processes associated with accumulated damage during aging and has been proposed as a therapeutic target for aging-related diseases, including neurodegenerative disorders [25]. AhR contributes to these diseases through proteins associated with cognitive decline and by mitigating neuroinflammation in the CNS [23, 26–28].

The AhR in Neuroinflammation Regulation

During aging, a chronic low-grade inflammatory state develops across various cells and tissues of the body, a phenomenon known as “inflammaging” [7]. This inflammatory state is linked closely to the senescence-associated secretory phenotype (SASP) [29]. Senescent cells do not proliferate and release cytokines and chemokines, creating an inflammatory, microenvironment that affects nearby cells [30]. This inflammatory environment contributes to disease susceptibility in multiple systems, including the CNS. In aging, neuroinflammation develops as a hallmark of neurodegenerative diseases [31, 32]. Glial cells such as astrocytes, oligodendrocytes, and microglia undergo senescence and contribute to neuroinflammation [33]. AhR exhibits ligand-dependent immunomodulatory functions, promoting or suppressing inflammatory responses [22].

AhR expressed in microglia, the resident immune cells of the CNS, inhibits inflammatory signaling and improves AD-associated memory deficits and behavior in transgenic models [34]. AhR expression is upregulated during LPS-induced inflammatory responses, consistent with its role in endogenous anti-inflammatory mechanisms. This is supported by preclinical evidence demonstrating that 6-formylindolo[3,2-b]carbazol (FICZ), an endogenous AhR ligand, attenuates inflammation in mouse models [35]. Furthermore, AhR

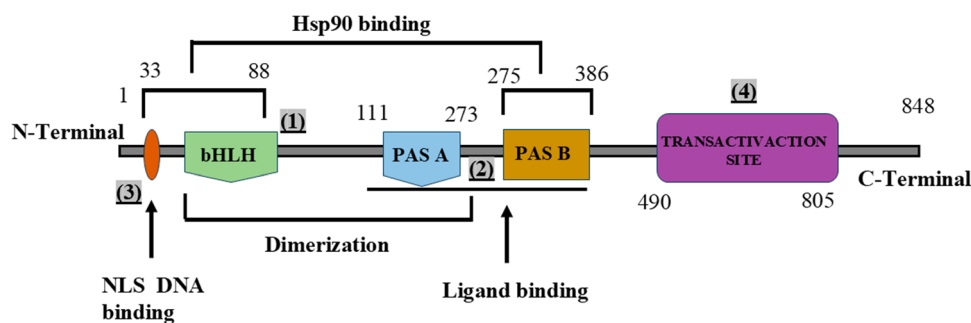


Fig. 2 Functional domains of the AhR and their molecular interactions. The AhR protein contains several critical domains that mediate its function: (1) bHLH domain facilitates DNA binding and dimerization with ARNT; (2) two PAS domains (PAS-A and PAS-B), where PAS-B specifically binds ligands (e.g., tryptophan metabolites) and interacts with Hsp90; (3) an NLS that directs nuclear translocation

upon activation; and (4) A transactivation domain that recruits transcriptional machinery. Hsp90 binds at the interface of the bHLH and PAS-B domains, maintaining the receptor in a cytosolic, inactive state until ligand binding triggers conformational changes, nuclear translocation, and subsequent gene regulation

activation in BV-2 microglial cells by the synthetic ligand 2,2'-aminophenylindole (2AI) significantly modulated inflammatory and oxidative stress responses, suppressing pro-inflammatory mediators (*iNOS*, *Ccl2*, *NLRP3*, *IL-1 β* , and *IL-6*) while upregulating antioxidant genes (*Nqo1*, *Hmox1*, and *Cat1*) [36].

In astrocytes, the primary glial cells responsible for neuronal support and CNS homeostasis, AhR mediates crosstalk with the immune system through STAT2-dependent type I interferon signaling [23]. Moreover, AhR activation in astrocytes contributes to the recovery of cognitive functions, stimulus discrimination, and motor functions in preclinical rodent models of neurodegenerative diseases, including AD, experimental autoimmune encephalomyelitis (EAE), and ALS [23, 37, 38].

AhR mediates anti-inflammatory responses by binding to xenobiotic response elements (XREs), thereby upregulating cytokines such as IL-10 and TGF- β [39, 40]. Additionally, AhR activation inhibits the NLRP3 inflammasome and NF- κ B signaling through either (1) competitive binding with p65/p50 subunits or (2) indirect induction of suppressor of cytokine signaling 2 (SOCS2) [23, 41].

Taken together, these findings support the idea that AhR is a relevant modulator of age-associated neuroinflammation; however, they do not yet justify a uniformly therapeutic interpretation. At present, the most defensible conclusion is that selective AhR modulation may prove beneficial in some inflammatory settings. Still, its consequences are likely to depend on ligand identity, exposure timing, target cell population, and the broader metabolic and immune context. This interpretive framework becomes especially important when considering gut microbiota as an upstream source of endogenous AhR ligands.

The GM-AhR Axis in Brain Aging and Neurodegenerative Disease

The GM typically maintains a homeostatic state known as eubiosis, and dysbiosis is defined as a disruptive alteration of the microbiota composition. It has emerged as both a consequence of age-related physiological decline and a significant risk factor for neurodegenerative disorders [4]. However, during aging, chronic inflammatory processes induce dysbiosis, leading to detrimental health effects [42, 43]. Dysbiosis triggers intestinal barrier hyperpermeability, leading to elevated proinflammatory cytokine levels, impaired nutrient absorption, and increased susceptibility to infection. These cumulative effects ultimately promote cognitive impairment in preclinical models [42, 44, 45].

AhR activation by bacterial tryptophan metabolites triggers signaling cascades with multifaceted neuroprotective outcomes, including promoting CNS cell proliferation and differentiation, strengthening tight junction protein expression, and

enabling β -amyloid (A β) clearance through metalloprotease induction, and balancing immune responses via concurrent anti-inflammatory cytokine upregulation and pro-inflammatory pathway suppression, as previously described [37, 38, 46].

The cognitive decline associated with aging primarily impacts processing speed, sustained attention, working memory, and cognitive flexibility, including task-switching and adaptive abilities, as well as higher-order executive functions such as learning and decision-making [47]. The GM-AhR axis serves as a critical regulatory interface in multiple physiological systems. Robust experimental evidence now establishes that AhR activation by microbial tryptophan metabolites constitutes a viable therapeutic approach for: aging-associated functional decline, affective disorders, and neurodevelopmental pathologies, including autism spectrum disorder (ASD), mediated through multitarget mechanisms [48].T. [49–51] (Fig. 3).

Current evidence reveals a significant gap in research addressing AhR modulation in non-pathological, age-related cognitive decline. While dietary interventions targeting GM have been extensively studied for their impact on cognitive aging in animal models, the clinical therapeutic potential of the GM-AhR axis remains underexplored in translational research contexts (Table 1).

Although AhR activation in aging-associated diseases has been examined mainly in preclinical models, human AhR shows higher affinity for indoles and greater involvement in cell proliferation and immune modulation [54]. In addition, peripheral administration of IPA increases its bioavailability in the brain and peripheral tissues and is associated with elevated levels of kynurenic acid, an endogenous AhR agonist [55].

Consistently, IPA reduces dysbiosis and enhances tight junction protein expression in animal models of non-alcoholic [56], reinforcing the relevance of the GM-AhR axis in aging-related processes and suggesting that low-dose activation may attenuate both primary aging mechanisms and cumulative damage.

Together, these findings underscore the need for well-designed clinical studies to characterize AhR signaling in humans, thereby validating the translational relevance of murine models of aging and neurogenesis.

AhR and Stem Cell Maintenance

The AhR plays a regulatory role in cell differentiation through modulation of key transcription factors, including Sox2, a critical regulator of pluripotency maintaining genes [57]. Importantly, AhR activation directly contributes to hippocampal neurogenesis and neuronal precursor differentiation [58]. Through the GM-AhR axis, microbial indoles orchestrate the β -catenin pathway, Neurog2,

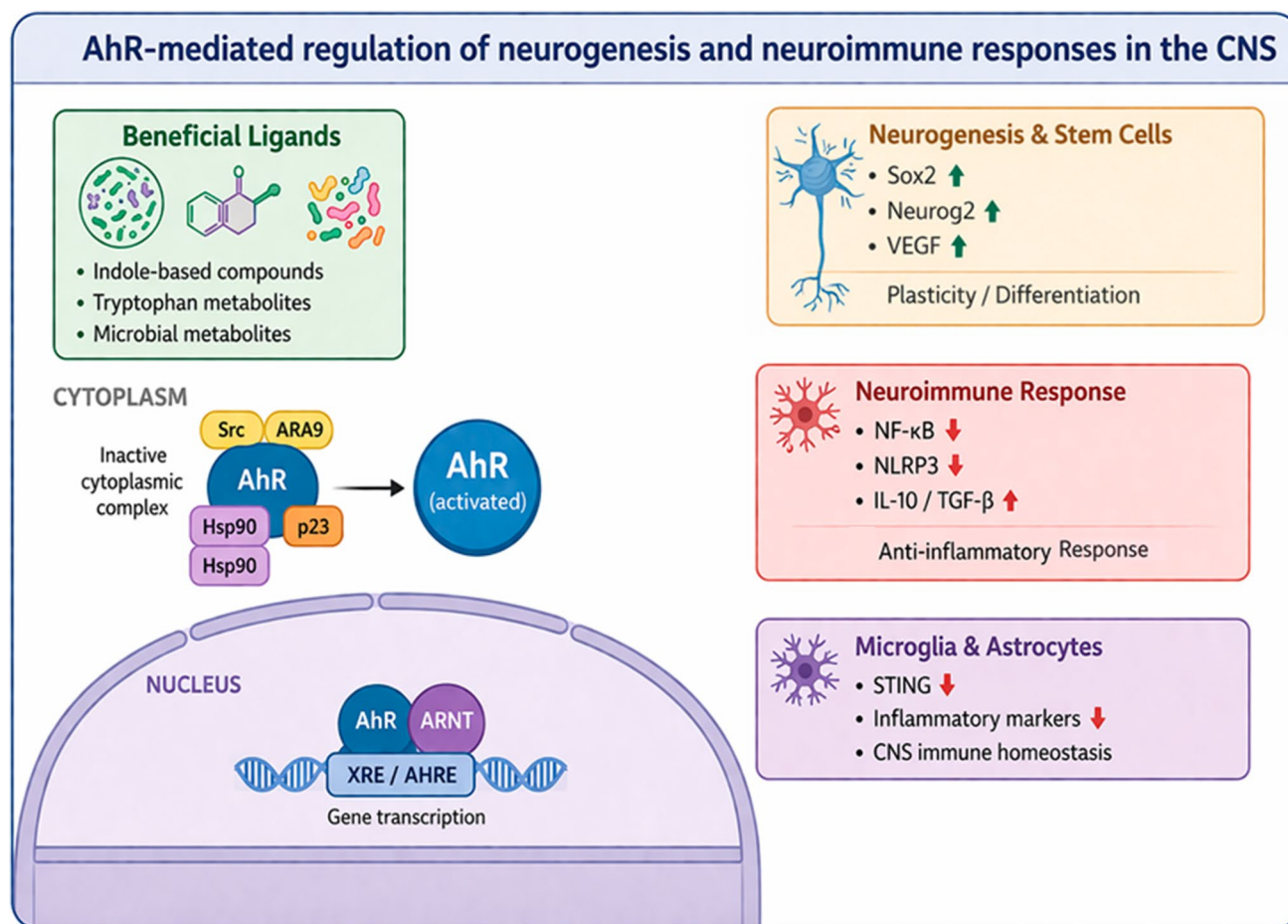


Fig. 3 Immunomodulatory and neurodevelopmental regulation mediated by microbiota-derived tryptophan metabolites via AhR activation. Tryptophan metabolites derived from the GM activate the AhR, promoting its nuclear translocation and initiating multiple anti-inflammatory pathways. These include direct inhibition of pro-inflammatory cytokine transcription through interaction with NF-κB or induction of anti-inflammatory mediators such as IL-10 and TGF-β. Together, these mechanisms constitute a coordinated anti-inflam-

matory response. In addition, AhR activation regulates developmental processes by modulating Sox2 expression and the expression of CNS development markers such as Neurog2 and VEGF. AhR also contributes to the maintenance of pluripotency in CNS cells, supporting their proliferation and subsequent differentiation, thus linking microbial metabolism to immunoregulation, stemness maintenance, and neural lineage specification

and VEGF signaling, collectively enhancing CNS cell proliferation and differentiation [59]. These mechanisms hold significant therapeutic potential for neurodegenerative disorders, including AD, PD, and ALS [52, 60–64]. Additionally, the interaction of activated AhR with Notch promotes increased production of neurotrophic factors during cell differentiation [63–65].

Stem cells, due to their ability to divide and differentiate into specific lineages, are essential in regenerative medicine and in the study of aging [4, 66]. In particular, neural stem cells can generate different types of central nervous system cells [67], a process that is fundamental for brain plasticity, which supports learning and adaptation but deteriorates with age and in neurodegenerative diseases [4]. Moreover, research shows that factors such as AhR activation and the cellular microenvironment influence this differentiation,

making them potential therapeutic targets for age-related disorders [66, 68].

Stem cells maintain prolonged activity throughout an organism's lifespan, during which they must cope with progressive accumulation of DNA damage. This persistent genomic stress can lead to instability that ultimately impairs the capacity for differentiation and compromises genetic integrity during aging.

The Ambivalent Role of AhR in Genomic Stability

The AhR plays a complex and sometimes context-dependent role in genomic instability, a hallmark of aging and neurodegenerative diseases. The endogenous AhR ligands,

Table 1 AhR activation mediates the modulation of neurological disease in experimental models by indole derivatives

Indole/indole derivative	Model system	Mechanism of action
Indole, indoxyl-3-sulfate (I3S)	Murine model of EAE. Primary cultures of astrocytes and microglia	Transcriptional regulation of AhR and reduction of inflammatory genes: NOS2, TNF- α , IL6, and CCL2 [23]
Indole propionic acid (IPA), indole-3-lactic acid (ILA), indole-3-carbinol (I3C), and I3S	Transgenic murine model of AD	Inhibition of the NLRP3 inflammasome, a decrease in proteins involved in the pathophysiology of AD, and restoration of cognitive processes via synaptic density proteins [34]
Indoles (unspecified)	Murine models (AhR $-/-$) and TnASA-deficient bacteria	The indole-AhR-mediated signaling pathway participates in hippocampal neurogenesis through the expression of the β -catenin, Neurog2, and VEGF genes [52]
IPA	A murine model of ASD is due to intrauterine growth restriction by caffeine	Activation of AhR by IPA increased social behaviors and reduced neuronal pruning by inhibiting NF- κ B activity in microglia [49, 50]
AhR agonists produced by GM (unspecified)	C57BL/6 J male mice	Modulation of Th and Treg cells via AhR reduced depressive symptoms by regulating these cells and decreasing neuroinflammation [17]
Indole-3-lactic acid (ILA)	5xFAD transgenic mouse model	AhR activation reduced amyloidopathy [46]
Indole-3-carboxaldehyde (3-IAld)	Mouse model for multiple sclerosis and EAE	Induces 5-HT secretion in mast cells, which promotes tolerance via Tregs cells [53]

tryptophan metabolites from the kynurenine pathway, affect the production of polymerases involved in translesion DNA synthesis, a cellular mechanism that allows DNA replication to continue despite damage, albeit at increased risk of errors [49, 50, 69]. This can increase genomic instability and, consequently, the probability of developing diseases, including neurodegenerative disorders [49, 50, 69, 70].

Building on this, elevated levels of endogenous AhR ligands, specifically tryptophan metabolites, have been reported in the cerebrospinal fluid of patients diagnosed with AD, PD, or ALS [71–74]. Additionally, an increase in γ -H2AX, a phosphorylated variant of histone H2A crucial for DNA organization, has been observed in neurodegenerative diseases [71, 75, 76]. γ -H2AX plays a key role in DNA damage response and genomic instability. Upon DNA damage, H2AX is phosphorylated to form γ -H2AX foci, which serve as signals to recruit repair proteins and complexes to the site of damage to facilitate correction. However, the elevated γ -H2AX levels observed in AD, PD, and ALS suggest heightened genomic instability resulting from impaired DNA repair mechanisms [71, 75, 76].

The AhR plays a significant yet complex role in DNA repair and maintenance of genomic stability. However, conflicting evidence exists: for instance, activation of AhR by TCDD has been shown to increase genomic instability, whereas resveratrol appears to mitigate DNA damage caused by environmental contaminants by competing with other AhR ligands [77].

These findings reveal complex regulatory dynamics in AhR signaling, in which ligand competition, potential allosteric interactions across multiple binding sites, and post-translational modifications collectively shape its functional outcomes. This multifaceted regulation likely underlies the receptor's context-dependent effects on genomic stability.

The AhR as a Regulator of Mitochondrial Homeostasis

Recent studies suggest that both activation and inhibition of AhR could be therapeutic targets for mitochondrial dysfunction, given its correlation with the mitophagy receptor BNIP3, a pro-apoptotic protein located in mitochondria. AhR inhibition leads to increased reactive oxygen species (ROS) levels and upregulation of BNIP3, whereas activation reduces BNIP3 expression and decreases the number of damaged mitochondria in primary mouse hepatocyte cultures [78].

In the CNS, inhibition of AhR has been shown to promote cognitive improvement in preclinical models of intracerebral hemorrhage, with similar effects observed after administration of an AhR antagonist, 6,2,4-trimethoxyflavone (TMF). In contrast, AhR activation by kynurenines increased mitochondrial ROS levels and hemorrhage in an AhR-dependent manner [79].

The involvement of the AhR in mitochondrial dysfunction remains debated. While some studies indicate that its activation worsens oxidative stress and compromises mitochondrial function [79], others propose context-dependent protective effects influenced by ligand specificity [12, 80, 81]. These opposing findings underscore the need to elucidate the precise molecular mechanisms through which this receptor modulates mitochondrial homeostasis.

Regulation of ATP production and Krebs cycle enzymes through AhR signaling helps prevent oxidative stress and maintain mitochondrial function, partially mediated via control of mitochondrial DNA biogenesis [82]. During senescence, decreased transcription of AhR target genes, such as CYP1A1 and CYP1B1, coincides with heightened cellular oxidative damage, suggesting that AhR activation could mobilize antioxidant defenses [81].

Despite clear links between mitochondrial dysfunction and age-related pathologies, research exploring AhR involvement in these conditions remains limited. This gap underscores the need for targeted investigations to determine whether modulating receptor activity could mitigate mitochondrial impairments in disease.

Antioxidant Mechanisms Mediated by AhR

Oxidative stress occurs when there is an imbalance between ROS production and the body's antioxidant capacity [83]. The AhR can activate protective pathways, such as NRF2; its context-dependent effects in aging, when antioxidant defenses are already diminished, require systematic evaluation. Studies suggest that oxidative stress plays a central role in the pathogenesis of neurodegenerative diseases such as AD, PD, and ALS, mainly through enhanced lipid peroxidation and reduced expression of antioxidant proteins [84–86].

AhR modulates oxidative stress by upregulating NRF2 expression through genomic pathways [87]. In immortalized human keratinocytes (HaCaT cells), oxidative stimuli increase AhR expression, which correlates with tryptophan metabolic activity and the formation of endogenous AhR agonists via oxidized tryptophan metabolites [88]. Furthermore, arachidonic acid-mediated AhR activation elevates ROS production through induction of cytochrome P450 (CYP) enzymes [89–91].

In the CNS, activation of AhR by β -naphthoflavone has been shown to increase the expression of antioxidant enzymes, including catalase (CAT) and glutathione S-transferases (GST), in pig brains [92]. These studies indicate a relationship between AhR, oxidative stress, and ROS production. Furthermore, they suggest the existence of bidirectional regulatory loops of potential biological and therapeutic interest.

Overall, the available evidence suggests that AhR functions as a fine-tuned modulator of cellular redox status rather than as a unidirectional regulator. Its involvement in ligand-dependent, tissue-dependent, and metabolic state-dependent feedback circuits positions AhR as an integrative node between environmental signals and adaptive responses. Within this framework, the functional interaction between AhR and NRF2 emerges as a potentially relevant mechanism for limiting the progressive cellular deterioration observed during senescence, although its net contribution appears to depend on specific physiological conditions and still requires systematic validation [93, 94].

The AhR Modulation in the Senescence

Lifestyle-related factors such as obesity and smoking have been associated with shorter telomeres in adult women, linking systemic inflammatory stress and accelerated aging [95]. Current evidence suggests that AhR may influence senescence in a biological response-dependent manner. On the one hand, impaired AhR homeostatic signaling has been proposed to favor senescence-associated inflammatory remodeling [96]. On the other hand, AhR activation by TCDD inhibited senescence in cultured keratinocytes [97], whereas the AhR agonist IPA enhanced macrophage phagocytic capacity, a process potentially relevant for senescent-cell clearance [98]. These findings support a cautious interpretation: AhR may affect both the induction and the tissue consequences of senescence, but direct mechanistic evidence in physiological brain aging remains limited.

AhR and Brain Lipid Metabolism: Implications for Myelination

The AhR is a key regulator of various endogenous metabolic pathways [99], acting as a constitutive inhibitor of triglyceride synthesis and an early regulator of adipocyte differentiation. It participates in lipid metabolism by regulating adipocyte function and lipid homeostasis [100].

In general, activated AhR promotes fatty acid oxidation, highlighting its metabolic regulatory function [89, 90, 101, 91]. However, when activated, it can also participate in phospholipid remodeling, in the synthesis of free fatty acids, and in triglyceride production [91].

There is growing evidence that AhR also acts as a sensor and regulator of lipid metabolism in the brain, influencing lipid homeostasis, protection, and stress response. During aging, AhR signaling becomes deregulated due to the accumulation of environmental toxins or changes in endogenous lipid metabolite levels [102], which can physiologically alter the functioning of cellular membranes and the formation of cholesterol- and

phospholipid-enriched lipid rafts, as well as communication and waste clearance through the formation of multivesicular bodies [103].

One of the important lipid-based structures of the nervous system is the myelin sheath. Structural alterations have been observed in the myelin sheath of the sciatic nerve in the peripheral nervous system and in the optic nerve of adult AhR $-/-$ mice. The elimination of the AhR receptor causes locomotor defects, thinning, and malformation of the myelin sheaths surrounding axons. One of the involved mechanisms is an increase in β -catenin levels due to AhR inhibition, which promotes its binding to myelin gene promoters [104, 105]. However, when AhR is activated, it also participates in regulating remyelination following damage-induced demyelination [104]. Moreover, glucocorticoid-induced AhR activation reduces inflammation and remyelination [106, 22], suggesting its therapeutic potential in ALS.

Aging decreases the availability of lipids required for myelin and reduces their renewal, leading to progressive demyelination, slower nerve conduction, and effects on functions such as memory and coordination [107]. However, the role

that AhR may play in this process remains unknown. It is interesting to discuss and investigate how exposure to toxic environmental pollutants that activate AhR may accelerate brain lipid damage and cognitive aging. In contrast, endogenous metabolites or dietary compounds that also act as AhR ligands [102], which may protect brain lipids and mitigate the effects of aging (Fig. 4). Taken together, the evidence suggests that AhR may play a dual role, both beneficial and detrimental, depending on the biological response to the different ligands analyzed. Because of this, it is of particular interest to further explore its protective role, so that this receptor may be considered a therapeutic target in demyelinating disorders.

The AhR as a Regulator of Key Aging Hallmarks and a Contributor to Neurodegenerative Processes

The antagonistic pleiotropy theory posits that certain genes regulate multiple beneficial functional traits during development but, in turn, produce harmful effects during aging. According to this theory, natural selection prioritizes

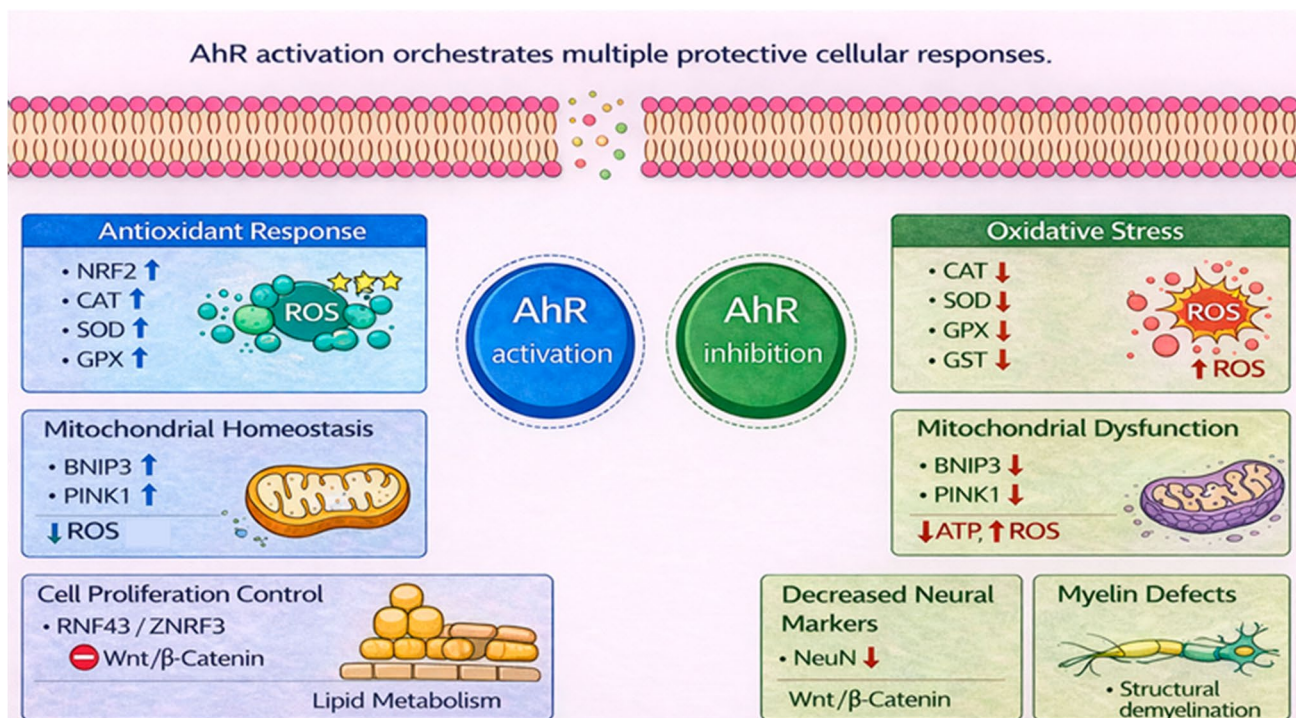


Fig. 4 The AhR mediates cellular response mechanisms. AhR activation orchestrates multiple protective cellular responses. (1) Under oxidative stress conditions, AhR induces the expression of NRF2, leading to the upregulation of key antioxidant enzymes such as catalase (CAT), superoxide dismutase (SOD), glutathione peroxidase (GPX), and glutathione S-transferase (GST), thereby reducing ROS and maintaining redox balance. (2) In terms of mitochondrial homeostasis, AhR activation increases the expression of BNIP3 and PINK1, both of which are essential for mitophagy and the preservation of

mitochondrial integrity. (3) AhR also induces the expression of RNF43 and ZNRF3, two E3 ubiquitin ligases that negatively regulate the Wnt/β-catenin signaling pathway, thereby preventing uncontrolled cell proliferation. (4) Conversely, AhR inhibition results in decreased BNIP3 expression, elevated ROS levels, mitochondrial dysfunction, reduced expression of neuronal markers, and structural alterations in optic nerve myelin, highlighting its neuroprotective role

youthful vitality over age-related frailty [108]. These genes exhibit a dual nature: they promote reproduction and growth in youth but lead to detrimental consequences later in life, thus driving the aging process [108]. Recent studies have expanded this theory by implicating the AhR functions in the regulation of multiple aging-related processes [109] (Fig. 4).

AhR plays essential roles during embryogenesis, particularly in stem cell pluripotency, cellular differentiation, and tissue morphogenesis, as described in earlier sections. However, its activity in adulthood has been linked to pathological aging processes, including vascular dysfunction, extracellular matrix degradation, and immunosenescence [96, 110].

Some authors attribute the shift in AhR functions to antagonistic pleiotropy, its multiple molecular targets, including aging, or the vast diversity of its ligands [96, 110]. However, changes in AhR function could also result from alterations in their expression, post-translational modifications that impair their genomic functions, or from unknown ligand dynamics yet to be discovered.

AhR as a Candidate Target and Biomarker in Aging Interventions

Several interventions commonly discussed in geroscience may modulate AhR signaling either by changing receptor expression or by altering the availability of endogenous ligands derived from the tryptophan-kynurenine axis. For example, analysis of aged mouse skeletal muscle transcriptomes showed that calorie restriction reduced AhR expression in selected muscles, whereas rapamycin did not significantly modify AhR expression in those datasets [111]. In humans, acute endurance exercise increased the availability of AhR ligands and activated AhR in peripheral blood mononuclear cells in an intensity-dependent manner [112]. These observations indicate that AhR is responsive to interventions relevant to aging biology, but they do not by themselves establish causal anti-aging efficacy through AhR modulation.

Ligand-Dependent Regulation of AhR Signaling

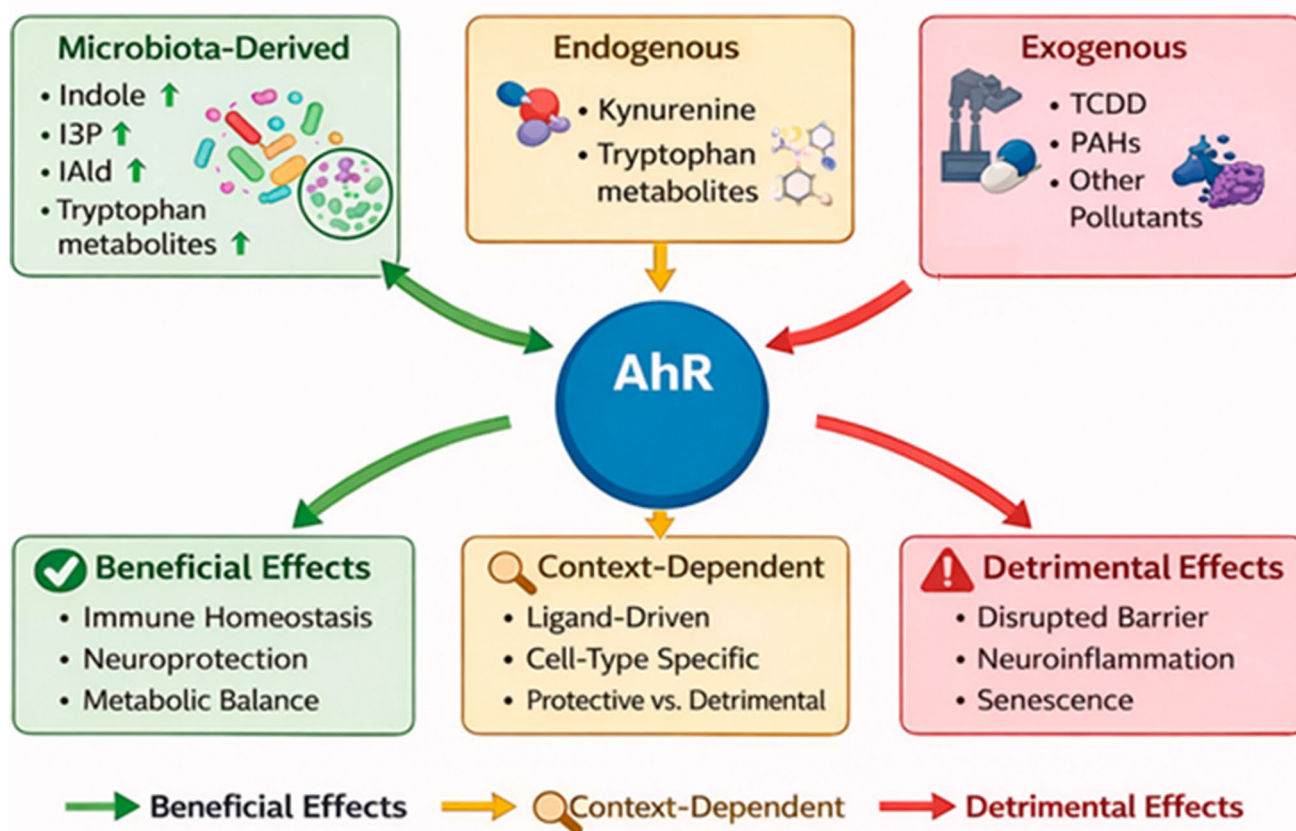


Fig. 5 Ligand-dependent and context-specific regulation of AhR signaling. AhR integrates signals from microbiota-derived, endogenous, and exogenous ligands, driving physiological responses. Depending on ligand type, cellular environment, and metabolic state, AhR activation can promote either beneficial effects (immune homeo-

stasis, neuroprotection, metabolic balance) or detrimental outcomes (barrier disruption, neuroinflammation, senescence). Thus, AhR functions as a ligand-sensitive regulator whose effects are determined by biological context

In neurological responses, this issue is particularly relevant because the kynurenine pathway generates endogenous AhR ligands and is tightly linked to inflammaging. Accordingly, interventions that shift kynurenine or kynurenic acid metabolism could influence brain immune and oxidative processes through the AhR axis [113]. Moreover, some putative geroprotective compounds have been reported to suppress or modulate AhR signaling. Resveratrol acts as a competitive AhR antagonist and prevents AhR-driven CYP1A1 induction in vivo [114], whereas metformin has been reported to downregulate AhR-responsive genes in cellular models [115]. Even so, the translational meaning of these observations remains uncertain because intervention studies rarely demonstrate that AhR modulation is the principal mechanism underlying the observed benefit.

AhR has also been proposed as a biomarker candidate because it intersects with processes relevant to brain aging, including adult hippocampal neurogenesis, neuronal maturation, neuroinflammation, and white matter integrity [58, 116]. However, this proposal should be treated with caution. AhR activity is highly dependent on tissue and cell type, ligand availability, microbiota composition, exposure history, comorbidities, and concurrent pharmacological treatments (Fig. 5). Consequently, an aging-related AhR readout may reflect environmental exposure, systemic inflammation, or disease state as much as chronological or biological aging itself.

For these reasons, the most realistic translational scenario is not the use of AhR as a stand-alone marker, but its incorporation into multi-parameter panels that combine receptor expression, target-gene signatures, metabolite profiles, and clinical metadata. Such an approach would better capture pathway activity while reducing the risk of overinterpreting a highly context-dependent signal [117, 118, 12, 94].

At present, AhR should be considered a plausible candidate target and a potentially informative pathway-level biomarker, but not yet a validated clinical marker of aging or a mature anti-aging intervention target.

Conclusions

In this review, we argue that AhR should be understood as a context-dependent regulator at the intersection of inflammatory, metabolic, redox, mitochondrial, microbiota-derived, and regenerative processes relevant to brain aging. The available evidence supports biological importance, but not a uniformly protective or deleterious interpretation. Instead, the consequences of AhR signaling appear to depend on ligand identity, exposure dynamics, cell type, and the distinction between physiological aging and established disease. This complexity helps explain why AhR emerges as both a promising therapeutic node and a difficult translational target.

Future progress will depend on moving beyond descriptive associations toward mechanistically resolved, cell-specific, and longitudinal human studies capable of defining when AhR modulation promotes resilience and when it may instead amplify aging-related dysfunction.

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Data Availability No datasets were generated or analyzed during the current study.

Declarations

Ethics Approval and Consent to Participate Not applicable.

Competing interests The authors declare no competing interests.

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